Perception-Integrated Execution in Robotic Culinary Environments

Autonomous Bimanual Fruit Skewering via Imitation Learning

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Challenge

Skewering food is a contact-rich task surprisingly difficult for low-cost robots. It requires precise coordination and the ability to handle soft, deformable objects like fruit.

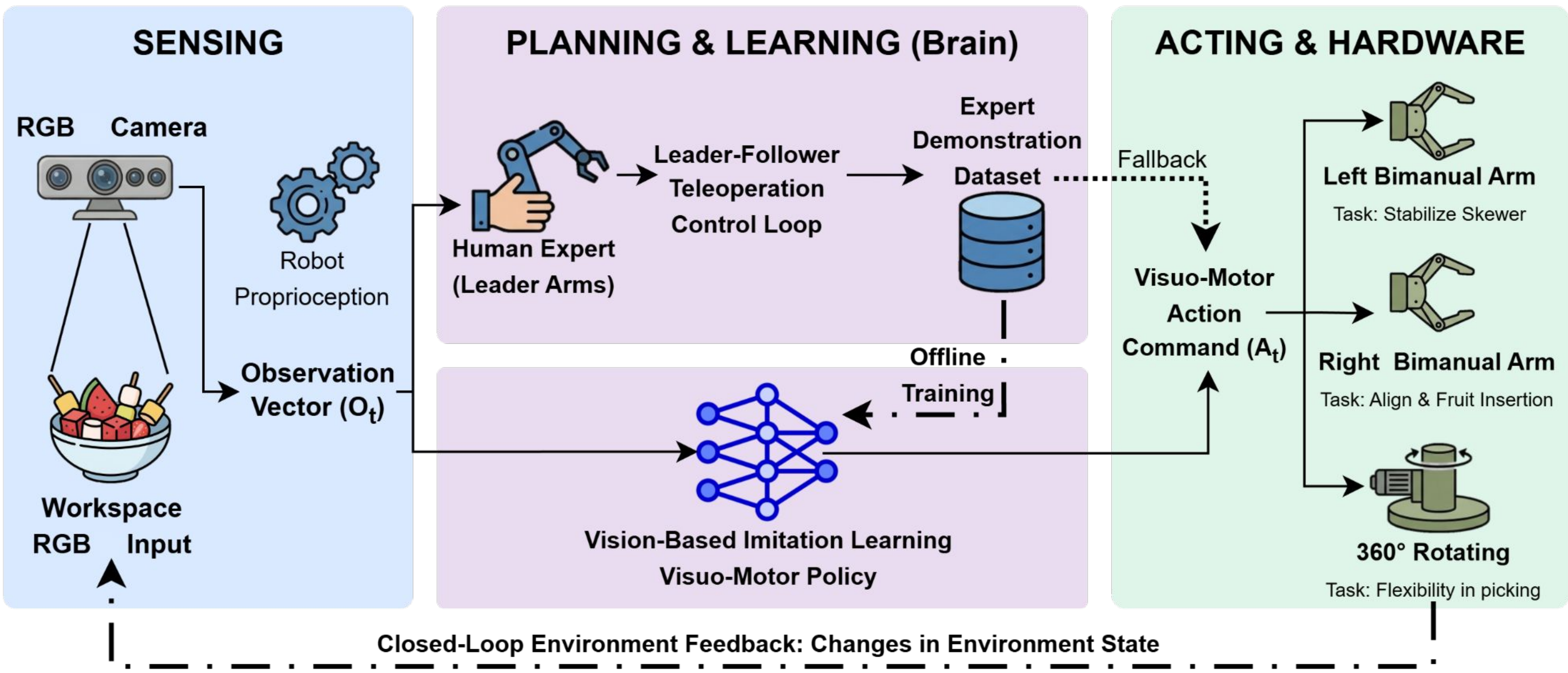
Solution

PIERCE learns complex bimanual skewering maneuvers through imitation learning. With 6 DoF and only visual feedback, it avoids expensive tactile sensors while maintaining high precision.

PIERCE can autonomously assemble a fruit skewer, and complete each of the following isolated actions:

- Detect objects and cups
- Pick up a single object
- Skewer an object
- Drop an object into a cup
- Sort different colored objects
- Human Interactions

SkewerBot System Architecture: SPA Loop & Imitation Learning Dataflow



Cross-Material Success: The Visuo-Motor policy successfully transferred from synthetic proxies (foam ball) to fruit with need for only minimal data.

Handling Uncertainty: Using a pretrained Vision-Language-Action (VLA) model allowed the robot to manage variations in object placement, orientation, and environment.

Sorting Capabilities: Beyond the primary skewering goal, the perception system demonstrated the ability to identify and organize objects based on visual attributes like color.

Evaluation Stage	Success Criteria	Result
Stage 1: Single-Item Grasping	Secure lift, no slipping or deforming	~ 90%
Stage 2: Insertion Precision	Accurate axial alignment, stable piercing	~ 85%
Stage 3: End-to-End Task	Autonomous assembly of 5-item skewer	~ 85%